PRD: Customer Churn Prediction

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Overview

This project builds a classification model that flags customers likely to churn and explains which factors drive that risk, so the business can act before customers leave.

Problem. Losing customers is costly, and the signals (falling usage, repeated support issues, short tenure) are scattered across usage and support data. Today nobody ranks customers by churn risk or knows which behaviours matter most.

Vision. A reproducible pipeline that goes from raw customer data to a ranked list of at-risk customers and a short list of business levers, backed by a clearly evaluated model.

Scope. Batch prediction on a historical dataset of customer usage with a churned / retained label. Real-time serving and automated retention campaigns are out of scope for v1.

Goals, non-goals and success metrics

The project succeeds if it ranks at-risk customers well enough to act on and tells the business what to change.

Goals

- Predict churn with a model that clearly beats a naive baseline on held-out data.
- Identify the behaviours and attributes most associated with churn.
- Deliver recommendations the business can act on, each tied to evidence.
- Keep the pipeline reproducible end to end.

Non-goals (v1)

- Real-time or API-based scoring.
- Causal claims; findings are associations, not proof that a change will reduce churn.
- Building the retention campaign itself.

Success metrics

Metric	Target	Why
ROC-AUC (test set)	At least 0.80	Ranking quality across all thresholds
Recall on churners	At least 0.70 at the chosen threshold	Missing a churner is the costly error
Precision on churners	At least 0.50 at the same threshold	Keeps outreach list worth acting on
Lift in top decile	At least 2.5x baseline churn rate	Shows the ranking is useful for targeting
Recommendations delivered	3 to 5, each backed by a chart or statistic	Makes the output actionable

Targets are starting points and should be revised once the real dataset's churn rate and size are known.

Users and stakeholders

Four groups use the project's output, and each needs a different view of it.

Stakeholder	Decision they make	What they need from the project
Customer success / retention team	Which customers to contact first	Ranked at-risk list with churn probability
Product and support leads	Which usage or service issues to fix	Top churn drivers with plain-language explanations
Business leadership	Where to invest to reduce churn	Short summary of findings and recommended focus areas
Project author (data scientist)	Which model and threshold to ship	Reproducible notebooks and a clear evaluation

Data requirements

The project needs one table with a row per customer, usage and service fields, and a churn label. Any public telco or SaaS churn dataset with this shape works; the real schema decides the final feature list.

Expected fields

Group	Example fields
Identity	customer_id (dropped before modelling)
Account	tenure in months, contract type, plan, payment method, monthly charges
Usage	logins or sessions per month, minutes or data used, features used, recent change in usage
Support	tickets raised, average resolution time, complaints
Label	churned (1) or retained (0)

Requirements

- Define churn explicitly (for example, cancelled within the observation window) and document it.
- Check duplicates, missing values, impossible values (negative tenure) and the class balance.
- Split into train, validation and test sets with stratification on the label; fix the random seed.
- Fit all encoders, scalers and imputers on training data only, to avoid leakage.
- Exclude any field recorded after the customer left, since it would leak the answer.

Functional requirements

The pipeline runs in five stages that mirror the project brief; each stage has a clear output.

1. **Load data.** Read the customer usage file, validate the schema, report row count, missing values and the churn rate. Output: a cleaned dataframe and a data-quality summary.

2. **Explore patterns.** Compare churned and retained customers on each field using summary statistics, distributions and churn rate by segment (tenure band, contract type, ticket count). Output: an EDA notebook with charts and a ranked list of candidate drivers.
3. **Engineer features.** Build tenure buckets, usage frequency and its recent trend, tickets per month, average resolution time, and charge-to-usage ratios; encode categoricals and scale numerics where the model needs it. Output: a documented feature table and a reusable preprocessing pipeline.
4. **Train models.** Fit a baseline (logistic regression), then tree-based models (random forest, gradient boosting) with cross-validated tuning and class-imbalance handling. Output: a saved best model and a comparison table.
5. **Evaluate and recommend.** Score the held-out test set, choose a decision threshold, explain drivers, and translate them into business actions. Output: an evaluation report and a one-page recommendations summary.

Modeling and evaluation plan

The model is chosen on validation performance and explainability, then judged once on the untouched test set.

Step	Approach
Baselines	Majority-class predictor and logistic regression
Candidate models	Random forest, gradient boosting (XGBoost or LightGBM)
Class imbalance	Class weights first; SMOTE or resampling only inside cross-validation folds
Tuning	Stratified 5-fold cross-validation with randomized search
Primary metrics	ROC-AUC and PR-AUC
Threshold metrics	Precision, recall, F1 and the confusion matrix at the chosen threshold
Business view	Lift and cumulative gains by decile

Step	Approach
Explainability	Feature importance, permutation importance and SHAP values

Choosing the threshold. Accuracy is misleading when churners are a minority. The threshold is set to hit the recall target while keeping precision acceptable, and the reasoning is documented so the business can move it as outreach capacity changes.

Turning results into advice. For each top driver, the report shows the direction of effect (for example, churn rate rises sharply with more than three tickets in 30 days) and a suggested action. These are associations from historical data and are labelled as such.

Deliverables and repository structure

The project ships five deliverables in one repository.

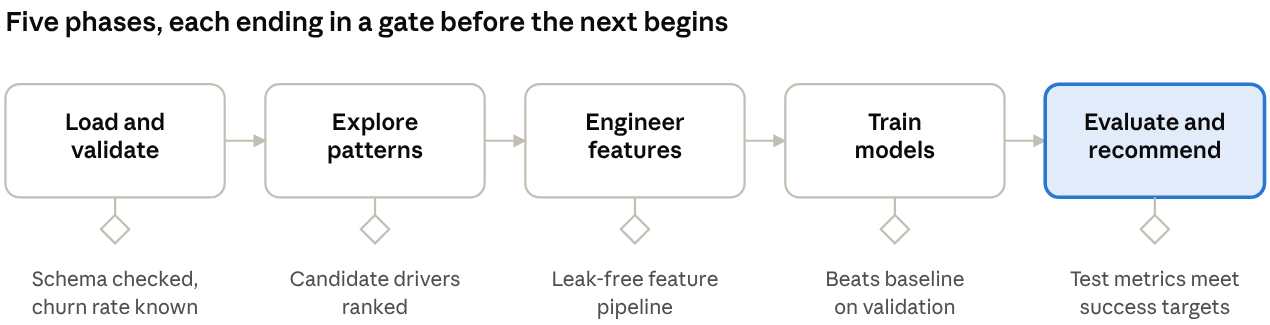
Deliverable	Format
Data-quality and EDA notebook	Jupyter notebook with charts
Feature engineering and training code	Python modules plus a notebook
Trained model and preprocessing pipeline	Saved artifact (joblib)
Evaluation report with SHAP explanations	Notebook or short PDF
Business recommendations summary	One page

```
churn-prediction/
  data/            raw and processed files (not committed if large)
  notebooks/       01_load_validate, 02_eda, 03_features, 04_model,
05_evaluate
  src/             data.py, features.py, train.py, evaluate.py
  models/          saved model and pipeline
  reports/         figures and recommendations summary
  requirements.txt
  README.md
```

Tooling. Python with pandas, scikit-learn, XGBoost or LightGBM, SHAP, matplotlib or seaborn.

Roadmap and milestones

The work runs in five phases that follow the pipeline, and no phase starts until the previous gate is met.



project roadmap · 5 phases, 5 gates

Each diamond marks the exit criterion for that phase. Calendar dates are left open until a start date and deadline are set.

Risks, assumptions and open questions

The biggest risks are data leakage and a label that does not match how the business defines churn.

Risk	Impact	Mitigation
Data leakage (post-churn fields, preprocessing on full data)	Inflated metrics that fail in practice	Review every field's timing; fit transforms on training data only
Severe class imbalance	Model ignores churners	Class weights, PR-AUC, recall-focused threshold
Overfitting on a small dataset	Poor generalisation	Cross-validation, simple baseline, held-out test set
Findings read as causal	Wasted retention spend	Label drivers as associations; suggest A/B tests
Poor data quality or missing support data	Weaker features	Document gaps; limit claims to available fields

Assumptions

- A single historical snapshot is enough; no time-series modelling in v1.
- Churn is a binary label already present in the dataset.

Open questions

- [ ] Which dataset will be used (public telco or SaaS data, or an internal one)?
- [ ] How is churn defined, and over what time window?
- [ ] Is recall or precision more important given outreach capacity?
- [ ] Is the deliverable a notebook report only, or also a small dashboard?